

ContextAgent

Offline, privacy-first accessibility for OpenHarmony.

Team: Context Window

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The Problem & The Solution

The Problem: Visually impaired users rely on robotic screen readers that lack context, and modern AI alternatives require streaming highly sensitive data (like medical bills or banking UI) to cloud servers. **Accessibility should not cost you your privacy.**

The Solution: ContextAgent is a privacy-first AI companion built natively for OpenHarmony. It runs a local Large Language Model (LLM) on-device to intelligently summarize the digital screen and physical environment with **zero cloud data leaks.**

Category Focus: Huawei (Imagine What's Next)

Native OpenHarmony Innovation

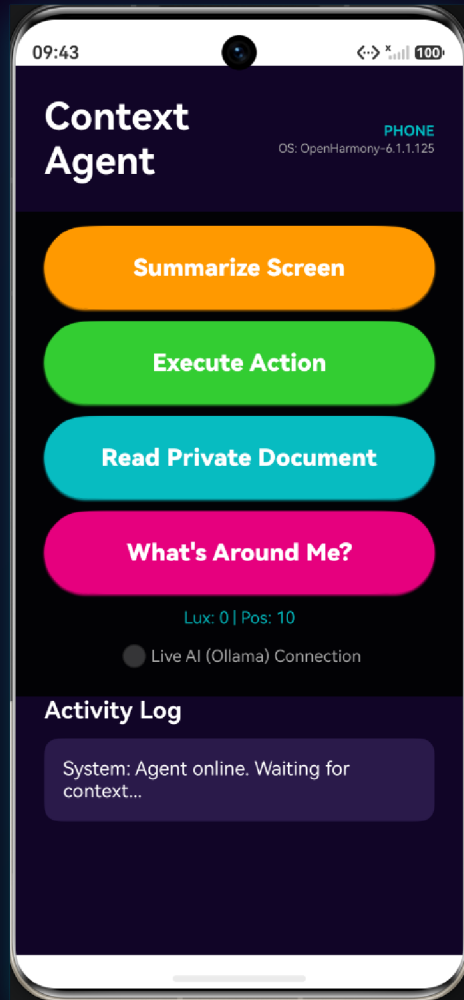
We didn't just build a standard web app. We built ContextAgent natively using **ArkUI** and **ArkTS** specifically to leverage OpenHarmony's native capabilities.

By utilizing the OpenHarmony HTTP network layer to communicate with a local LLM, and hooking directly into the native `@ohos.sensor` APIs, we bridge the gap between digital screen reading and physical hardware awareness.

How It Works (The Tech)

1. **Local AI Engine:** Powered by a local **Ollama (Gemma 4)** instance bridged to the OpenHarmony app.
2. **Digital Context:** Instead of reading every button line-by-line, the LLM generates a natural conversational summary.
3. **Hardware Awareness:** Native OpenHarmony sensors provide real-time Gravity (Y/Z) and Ambient Light (Lux) data to the AI.
4. **Conversational State Machine:** Complex actions (like banking transfers) are safely executed via step-by-step voice confirmation.

The ContextAgent Dashboard



Feature 1: The Digital Screen Reader

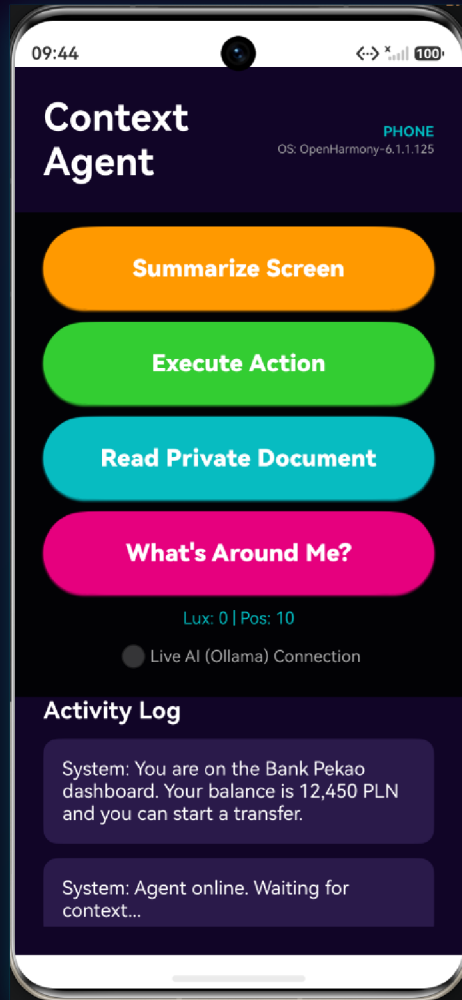
Traditional screen readers blindly read everything: "Image 1. Button. Header. Balance 12,450. Transfer Button."

ContextAgent understands intent.

When a user opens the Bank Pekao app, the AI contextually summarizes the page:

"You are on the Bank Pekao dashboard. Your balance is 12,450 PLN and you can start a transfer."

Contextual Screen Summarization

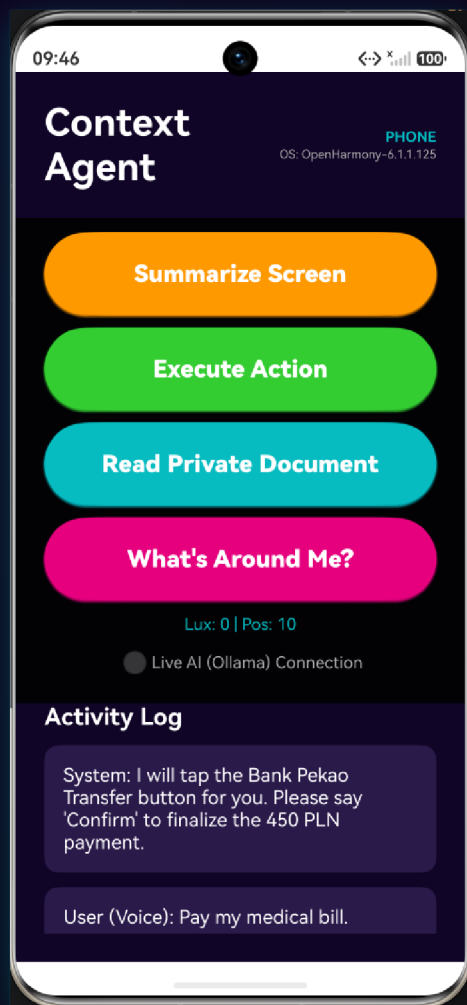


Feature 2: Agentic Action (Bank Pekao)

But reading isn't enough—the agent can securely act on the user's behalf using voice commands.

1. User says: **"Pay my medical bill."**
2. The AI securely analyzes the UI and prepares the exact 450 PLN transaction.
3. **Safety First:** It pauses and asks the user: **"I will tap the Bank Pekao Transfer button for you. Please say 'Confirm' to finalize."**
4. User confirms, and the native action executes.

Safety & Action Confirmation



Feature 3: Hardware Awareness

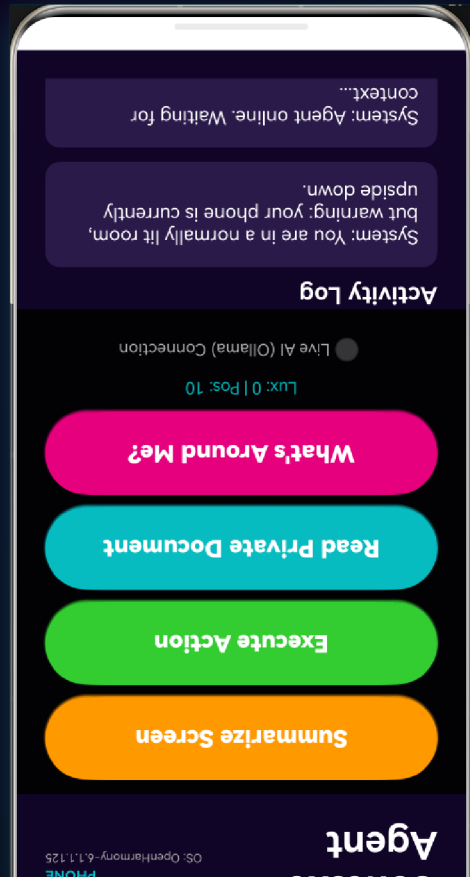
Because ContextAgent is native to OpenHarmony, it is physically aware of the device.

We mapped the `@ohos.sensor` **Accelerometer** and **Ambient Light** sensors directly to the AI's prompt context.

If a visually impaired user accidentally holds their phone upside down, the agent instantly detects the gravity shift (`gravityY > 5`) and triggers an auditory warning:

"You are in a normally lit room, but warning: your phone is currently upside down."

Native Sensor Integration (@ohos.sensor)



Summary & Links

Powering the next generation of privacy-first accessibility.

Demo Video:

<https://youtube.com/shorts/d7PVtOz6WnE>

Code Repository:

<https://github.com/thoufeeque22/ContextAgent-OpenHarmony>

ContextAgent by Team Context Window